THE BENEDICT YEAR

Science from final causes

Start where ends are plain, in things people make, and only then turn to things that grow, whose ends are their own.

Richard Verver, JCL · Canon Lawyer

Science starts where ends are plain, in things people make, and only then turns to things that grow, whose ends are their own. That is Aristotle's own route. It keeps two errors off the table: mechanism, for which nothing is for anything, and Intelligent Design, which treats an organism as a watch with its designer outside it.

The four causes, on a made thing

Aristotle's example is the essays' Making page's own: the bronze of the statue (*Physics* II.3, 194b23). Its matter is bronze, its form the shape, its maker the sculptor, and its end what it is for. Mortimer Adler's *Aristotle for Everybody* (1978) teaches them the same way. In its part "Man the Maker" he asks a shoemaker, a jeweller and a gunsmith about their work. He wrote the book for readers "from twelve or fourteen years old upward".

Then nature

An artifact's end is put on it from outside. A natural thing has its end within: nature is an inner principle of motion and rest (II.1, 192b). Art imitates nature (II.2, 194a21; II.8, 199a), so ends are learned on art first, and then nature is seen not to be art.

That is where Edward Feser draws the line against Intelligent Design: Paley-style design arguments "tend to assimilate natural substances to artifacts" (*Aristotle's Revenge*, 2019). That God is the source of nature's order is a further step, the Fifth Way, not a finding. Not every Thomist draws the line as sharply; Marie George is one who does not.

The course: *Caveman Chemistry*

Kevin Dunn's book (2003) is 28 projects in historical order, "from the creation of fire to the production of plastics". He wrote it for his course at Hampden-Sydney College, and the whole text is free at cavemanchemistry.com. Each project starts from a thing people needed, which puts the end first. It is to chemistry what *NAND to Tetris* is to computers: the whole built up from first things, in order.

PROJECT	WHAT IT IS FOR	UNIT
1 Lucifer (Charcoal)	A hotter, cleaner fire	4 Making
3 Hammurabi (Units)	Fair measure	2 Number
5 Athanor (Ceramics)	Vessels that hold	4 Making
13 Theophilus (Glass)	Glass; Theophilus was the monk whose craft book covers church windows	12 Worship
14 Ts'ai Lun (Paper)	Something to write on	1 Reading
21 Volta (Batteries)	Stored current. Its electrode is charcoal, so make project 1 first, or buy it	3 Nature
23 Eastman (Photography)	Fixing an image	4 Making

Safety. It is a college lab book, and every project's notebook entry carries a safety section. Dunn himself says most people lack the judgment to do the gunpowder project safely. For a child of 12 or 13 at home:

- **With an adult:** charcoal, ceramics, textiles, paper, natural dyes (project 12, not the aniline dyes of project 22), and soap, whose lye needs close supervision.
- **With an adult and protection:** glass (a kiln, a dust mask, glasses); batteries (an alkaline solution that gives off hydrogen, and Dunn treats an eye splash as an emergency); photography (silver nitrate: gloves and glasses).
- **Read, but do not do:** gunpowder, acid, soda and chlorine.
- **Adults only:** mead and alcohol.

A school science credit

In Ontario, the de-streamed Grade 9 science course (SNC1W) has five strands: STEM skills; biology (ecosystems and climate change); chemistry (the nature of matter); physics (electricity); and earth and space science (space exploration). It can be held two ways:

- **Keep it for the credit**, read through the four causes, with its chemistry strand (atomic structure, Bohr-Rutherford diagrams, the periodic table) laid on the crafts as findings. Its own expectation C2.2, on how experimental evidence shaped models of the atom, is the Nature unit's rule at work: a finding fixes a fact until a later finding replaces it.
- **Let the credit wait**, make *Caveman Chemistry* the year's science, and take the course later; TVO's Independent Learning Centre offers it as independent study. The diploma asks for two science credits,

not Grade 9 science as such, but Grade 10 academic and applied science list it as a prerequisite, though a principal can waive one.

Big History, and what it can say

The Big History Project, free from the OER Project and built for middle school, runs through eight thresholds of increasing complexity, from the Big Bang to the modern revolution; Rise Out sells a homeschool version in two self-paced parts, from the cosmos to agriculture, for 13 and up. It is Montessori's Great Stories grown up, and it carries Nature and Memory. Two cautions keep it to what it can say: its findings are dated and replaceable, and a story of complexity tells about the parts, never what a thing is or what it is for. In 5 Memory the child lays its thresholds beside the atrium's ribbon of the History of the Kingdom of God: one order of events, found and taught, and only the found part carries dates.

Next year

NAND to Tetris Part I needs no programming and builds a computer up from a single logic gate. It is free, and computer science lives in Number and Making on the essays' map.

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